

Protein Structure Analysis Using PyMOL

Proteins

Proteins are polymers of amino acids. Amino acids are connected through peptide bonds, forming a polypeptide chain. The functional properties of proteins depends upon their three dimensional structures.

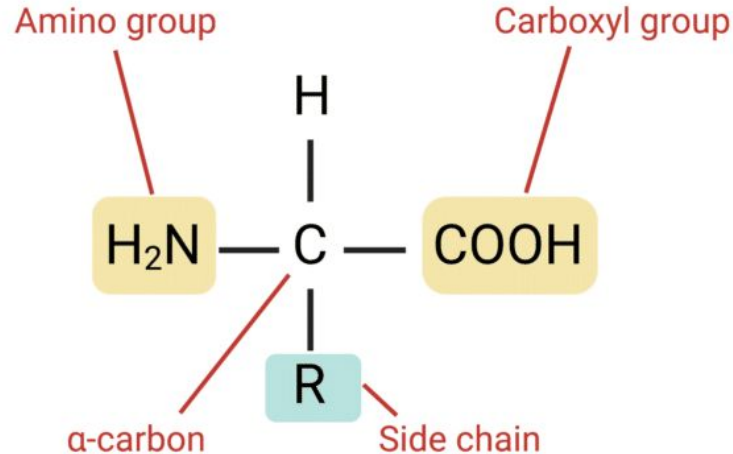
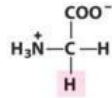


Figure: Structure of an amino acid

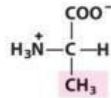
The 20 Amino Acids Found in Proteins

R-groups

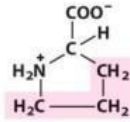
Nonpolar, aliphatic R groups



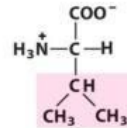
Glycine



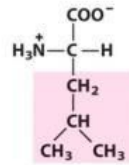
Alanine



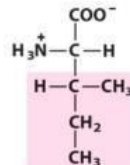
Proline



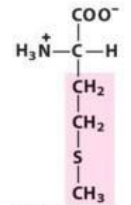
Valine



Leucine

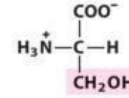


Isoleucine

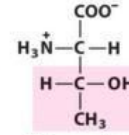


Methionine

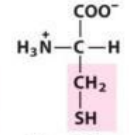
Polar, uncharged R groups



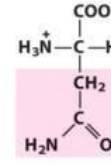
Serine



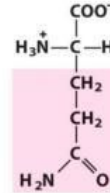
Threonine



Cysteine

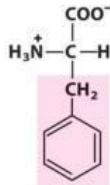


Asparagine

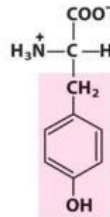


Glutamine

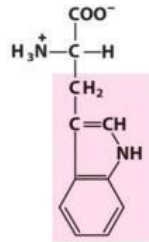
Aromatic R groups



Phenylalanine

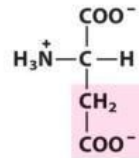


Tyrosine

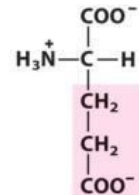


Tryptophan

Negatively charged R groups

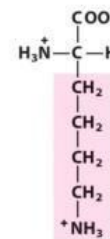


Aspartate

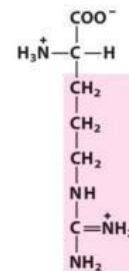


Glutamate

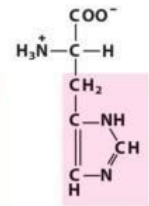
Positively charged R groups



Lysine



Arginine



Histidine

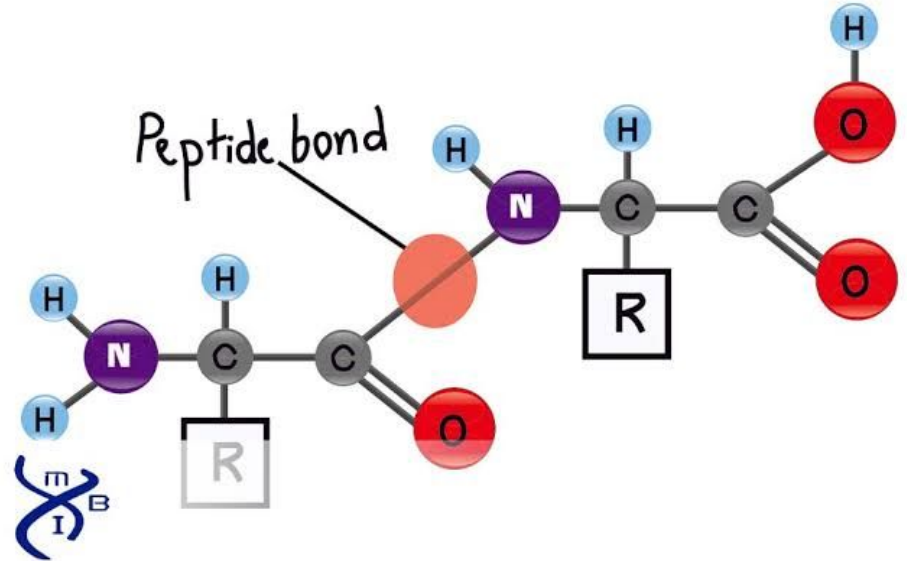
Peptide Bond

Amino acids in polypeptide chain are linked by peptide bonds.

In polypeptide chain, the carboxyl group of n th amino acid forms a peptide bond ($C\alpha-N$) with amino group of $(n+1)$ th amino acid.

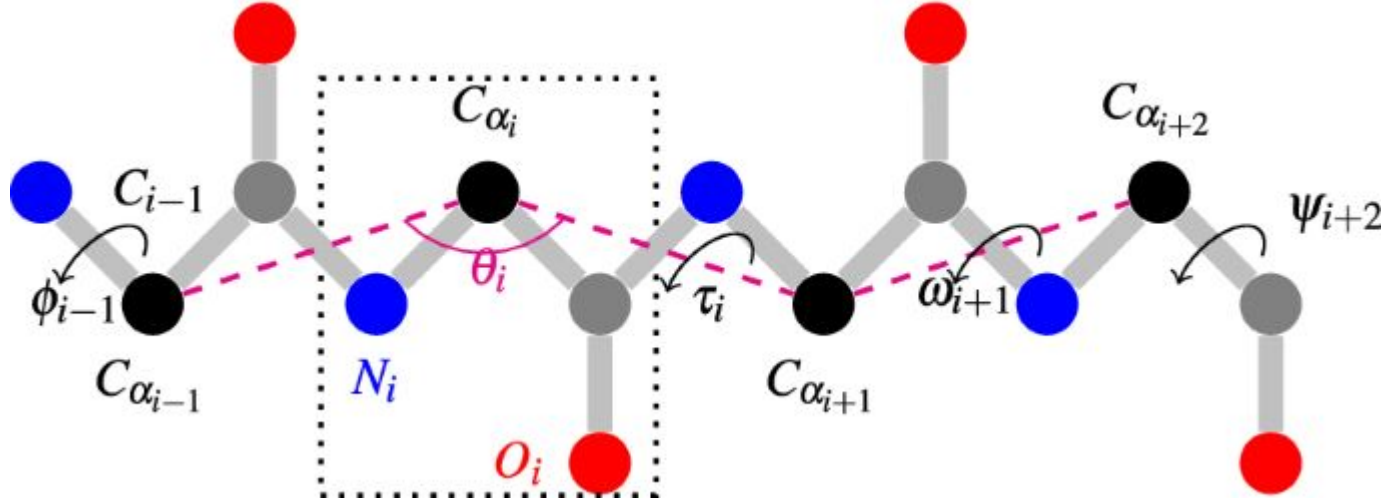
When peptide bond forms, water molecule is released and what remains of each amino acid is called as an amino acid residue.

Residues differ at only side chain

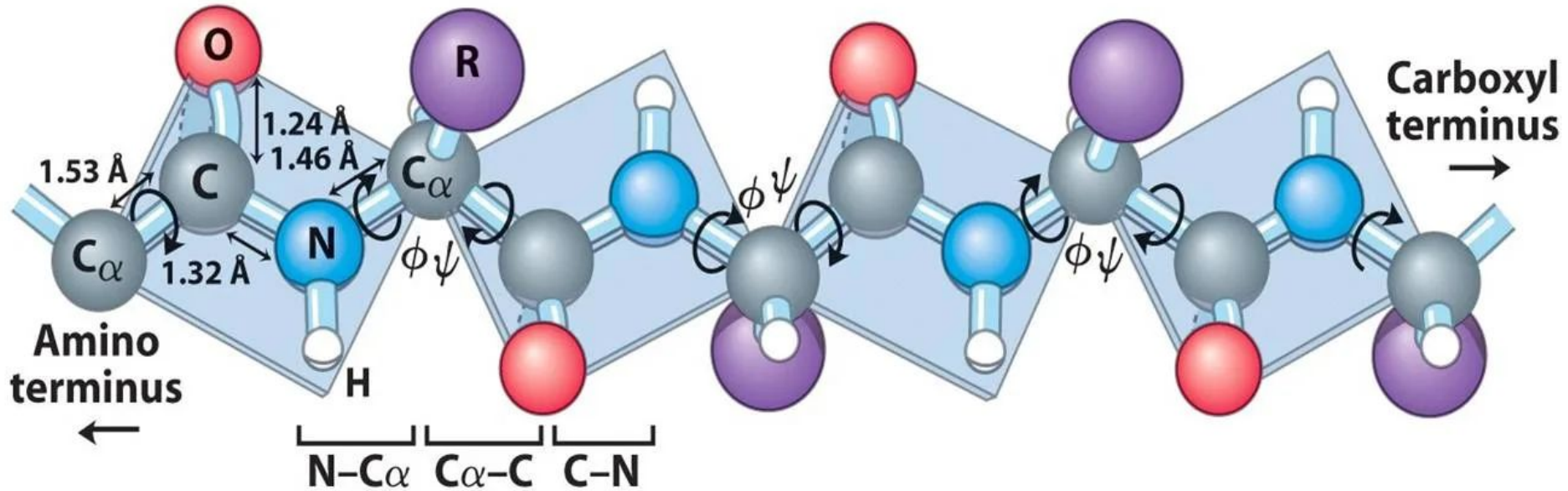


Peptide Chain and Protein Backbone

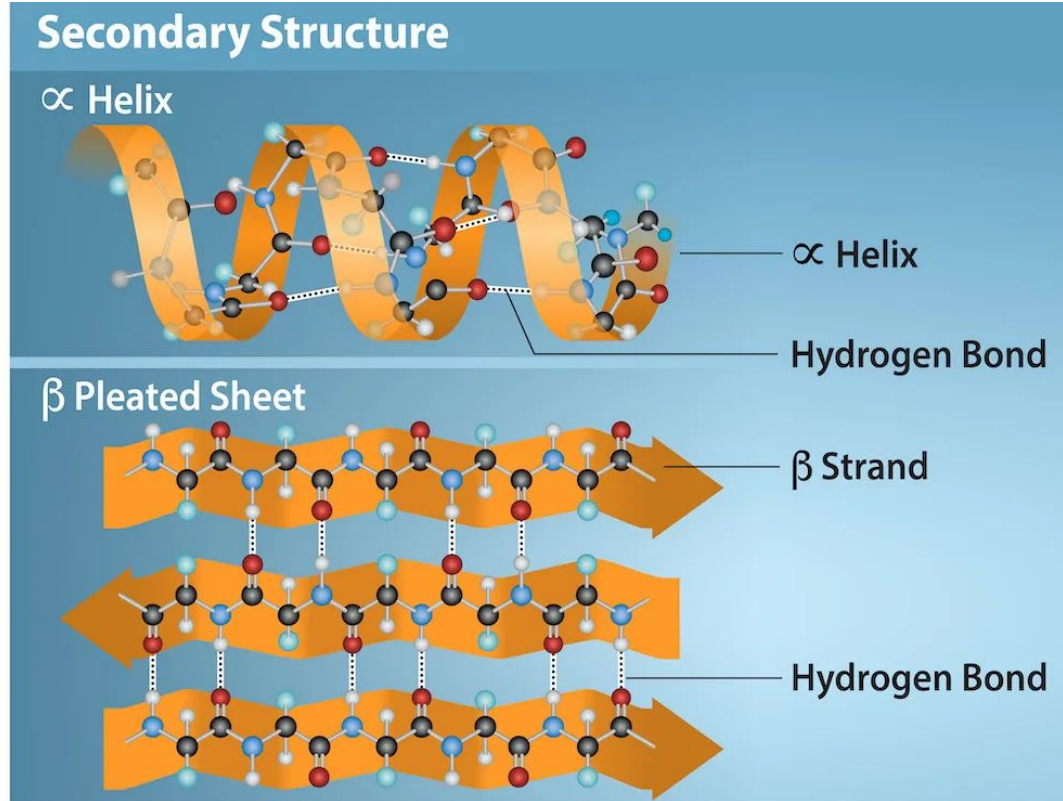
Once many amino acids are connected by peptide bonds, they form a polypeptide chain that becomes the basis of a protein. Although the chain contains many different amino acids and side chains, it has a repeating protein backbone made up of the atoms N-C α -C. The R-groups (side chains) extend from the C α atoms and are what distinguish one amino acid from another.

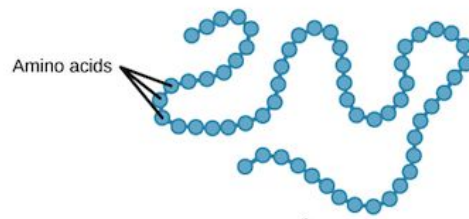


Peptide plane

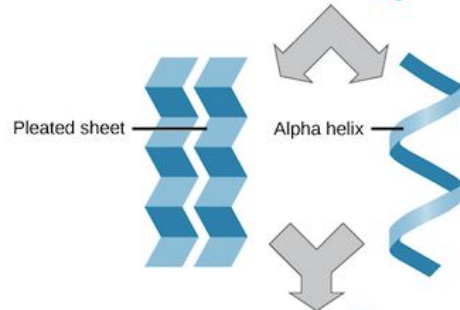


Secondary Structures





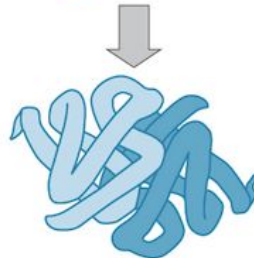
Primary protein structure
sequence of a chain of
amino acids



Secondary protein structure
hydrogen bonding of the peptide
backbone causes the amino
acids to fold into a repeating
pattern



Tertiary protein structure
three-dimensional folding
pattern of a protein due to side
chain interactions



Quaternary protein structure
protein consisting of more
than one amino acid chain

Protein Data Bank (PDB) (<https://www.rcsb.org/>)

The Protein Data Bank is a database for the three-dimensional structural data of large biological molecules, such as proteins and nucleic acids.

The screenshot shows the top navigation bar of the RCSB PDB website. It includes a dark blue header with white text for navigation: RCSB PDB, Deposit, Search, Visualize, Analyze, Download, Learn, About, Careers, and COVID-19. On the right are buttons for Help, Contact us, and MyPDB. Below the header is a white section with the PDB logo, statistics (259,412 Structures from the PDB archive, 1,062,058 Computed Structure Models (CSM)), a search bar with the placeholder 'Enter search term(s), Ligand ID or sequence', and an 'Include CSM' checkbox. A 'Help' link is on the far right. A secondary bar contains logos for PDB-101, PDB, EMDDataResource, NAKB, wwPDB Foundation, and PDB-IHM, along with social media icons for Facebook, X, YouTube, and LinkedIn.

On July 21, 2027, wwPDB will fully transition to extended 12-character PDB IDs, the PDBx/mmCIF format, and the re-organized PDB archive.

Start using the **PDB Beta Archive** today.

The screenshot shows the main content area of the RCSB PDB website. On the left is a dark blue sidebar with white text and icons for: Welcome, Deposit, Search, Visualize, Analyze, Download, and Learn. The main content area has a white background. At the top, a text block states: 'RCSB Protein Data Bank (RCSB PDB) enables breakthroughs in science and education by providing access and tools for exploration, visualization, and analysis of:'. Below this are three items with icons: 'Experimentally-determined 3D structures from the Protein Data Bank (PDB) archive', 'Integrative 3D Structures from the PDB Archive', and 'Computed Structure Models (CSM) from AlphaFold DB and ModelArchive'. Below these are two promotional banners: 'NEW Explore Integrative Structures' featuring a puzzle piece icon, and 'PDB-101 Training Resources' featuring a lightbulb icon with a 'PDB-101' tag. On the right is a section titled 'September Molecule of the Month' with a 3D molecular model of a protein complex (Biofilms) and a yellow ribbon. The word 'Biofilms' is written below the model.



Structure Summary

Structure

Annotations

Experiment

Sequence

Genome

Versions

Biological Assembly 1 ⓘ

Explore in 3D: [Structure](#) | [Sequence Annotations](#) | [Validation Report](#)

Global Symmetry: Cyclic - C2 ⓘ ([Explore in 3D](#))

Global Stoichiometry: Homo 2-mer - A2 ⓘ

[Find Similar Assemblies](#)

Biological assembly 1 assigned by authors and generated by PISA,PQS (software)

Macromolecule Content ⓘ

- Total Structure Weight: 35.49 kDa ⓘ
- Atom Count: 2,568 ⓘ
- Modeled Residue Count: 318 ⓘ
- Deposited Residue Count: 327 ⓘ
- Unique protein chains: 1

1PTA | pdb_00001pta ⓘ

THREE-DIMENSIONAL STRUCTURE OF PHOSPHOTRIESTERASE: AN DETOXYIFYING ORGANOPHOSPHATE NERVE AGENTS

PDB DOI: <https://doi.org/10.2210/pdb1PTA/pdb>

Classification: [HYDROLASE \(ORGANOPHOSPHATE-DEGRADING\)](#)

Organism(s): [Brevundimonas diminuta](#)

Mutation(s): No ⓘ

Deposited: 1994-07-07 **Released:** 1995-12-01

Deposition Author(s): [Benning, M.](#), [Holden, H.M.](#)

Experimental Data Snapshot

Method: X-RAY DIFFRACTION

Resolution: 2.10 Å

R-Value Observed:

0.180 (Depositor) ⓘ

Display Files ▾

Download Files ▾

Data API

FASTA Sequence

PDBx/mmCIF Format

PDBx/mmCIF Format (gz)

BinaryCIF Format (gz)

Legacy PDB Format

Legacy PDB Format (gz)

PDBML/XML Format (gz)

Validation Full (PDF - gz)

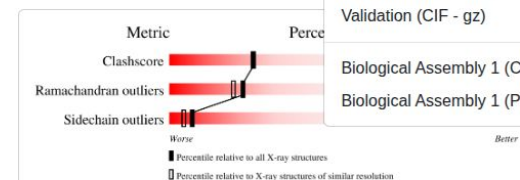
Validation (XML - gz)

Validation (CIF - gz)

Biological Assembly 1 (CIF - gz) ⓘ

Biological Assembly 1 (PDB - gz)

wwPDB Validation ⓘ



This is version 1.4 of the entry. See complete [history](#).

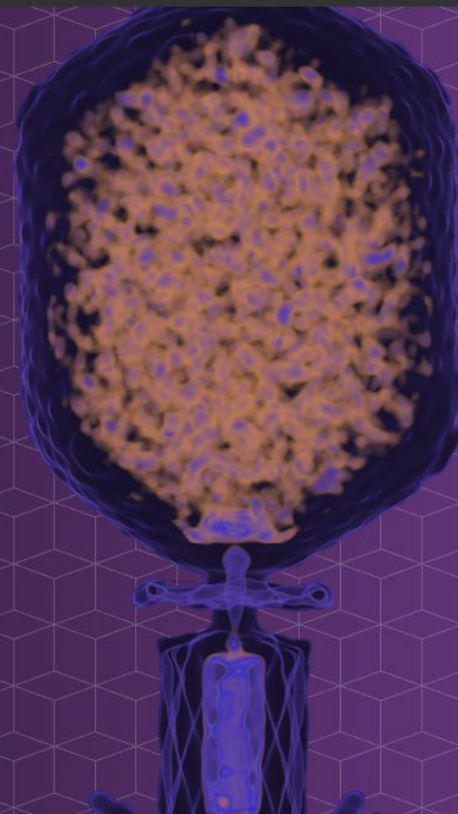
Introducing PyMOL 3.1

PyMOL is a user-sponsored molecular visualization system on an [open-source foundation](#), maintained and distributed by [Schrödinger](#).

[DOWNLOAD NOW](#)

[BUY LICENSE](#)

[PYMOL 3 PRODUCT PAGE](#)



/1PTA/A/A/36 41 46 51 56 76 94 101 106 111 116 121 126 131 136 141 146 151 156 161 166 171
QINTVRGPITISEAGFTLTHEICG-----R-----GVRTIVDVSTFDIGRDVSLAEVSRADVHIVAATGLWFDPPLSMRLRSVEELTQFFLREIQYGIEDTGIRAGIIKVATTG

No License File - For Evaluation Only (30 days remaining)



- All A S H L C
- 1PTA A S H L C
- (sele) A S H L C
- (res20) A S H L C
- (res36) A S H L C

3-Button Viewing SEQ

```
THR 361
LEU 362
Iterate: iterated over 327 atoms.
PyMOL>editing_ring cut
PyMOL>select res36, chain A and resi 36
Selector: selection "res36" defined with 11 atoms.
PyMOL>stored.residues = []
PyMOL>iterate polymer.protein and name CA, stored.residues.append(resn)
Iterate: iterated over 327 atoms.
PyMOL>print(sorted(set(stored.residues)))
['ALA', 'ARG', 'ASN', 'ASP', 'CYS', 'GLN', 'GLU', 'GLY', 'HIS', 'ILE', 'LEU', 'LYS', 'MET', 'PHE', 'PRO', 'SER', 'THR', 'TRP', 'TYR', 'VAL']
PyMOL>show sticks, res36
You clicked /1PTA/A/A/LYS`339/CA
Selector: selection "res36" defined with 20 atoms.
```

Commands

Load & inspect

```
load protein.pdb  
get_chains all  
set seq_view, 1  
iterate polymer.protein and name CA, print(resi, resn, ss)
```

Visualize

```
hide everything  
show cartoon  
show sticks  
orient  
zoom  
bg_color white
```

Commands

Select residues

select res1, chain A and resi 45

show sticks, res1

zoom res1

Label backbone atoms

label res1 and name N+CA+C, name

Select R-group

select sc1, res1 and not name N+CA+C+O

show sticks, sc1

Hydrogen-bond distance

distance hb1, /PDB//A/102/N, /PDB//A/106/O

Calculate ϕ (phi)

get_dihedral /PDB//A/101/C, /PDB//A/102/N, /PDB//A/102/CA,
/PDB//A/102/C

Calculate ψ (psi)

get_dihedral /PDB//A/102/N, /PDB//A/102/CA, /PDB//A/102/C,
/PDB//A/103/N

Save image

png my_structure.png, dpi=300

Questions

1) Load the PDB structure assigned to you in PyMOL and examine its overall 3D structure.

a. Note the following information:

Protein name, PDB ID, Number of chains present in the structure

b. Display the protein using cartoon representation.

c. Briefly describe the overall shape of the protein. Is it relatively compact, elongated, globular, etc.?

d. Take a screenshot of the complete protein structure in PyMOL.

Q2. Explore the Amino Acids and Their R-Groups

Select five different amino-acid residues from your protein and display them using ball-and-stick/stick representation. And for selected residue report Residue, Residue No, R-group type.

- a. Which atoms form the protein backbone of each residue?
- b. Which part represents the R-group/side chain?
- c. Identify at least:
 - one hydrophobic residue
 - one polar residue
 - one charged residue

Questions

Q3 Identify the N, C α and C atoms and the R-group attached to the C α of each of the three selected residues, display the residues in stick representation, take a screenshot, and describe what remains the same and what changes between amino-acid residues.

Q4. Show the hydrogen bond in any one alpha-helix, parallel beta-strands, and antiparallel beta strands of the given protein. Also, calculate their average hydrogen bond length.

Q5 Find a β -sheet and identify two neighboring β -strands; determine whether they are parallel or antiparallel, then identify and measure the distance of one hydrogen bond between them.

Q6 Select one residue each from an α -helix, β -strand, and loop, and use PyMOL to determine their ϕ (phi) and ψ (psi) angles.

Q7 Compare the three sets of ϕ/ψ values: how do they differ between secondary structures, which shows the greatest backbone conformational difference, and why?

Q8 Build a Ramachandran Plot

Using the ϕ and ψ values obtained from your protein, construct a Ramachandran plot.

Create separate plots/markings for:

1. α -helix residues
2. β -strand residues
3. Loop residues

Interpretation

- a. Where do the α -helix residues tend to cluster?
- b. Where do the β -strand residues tend to cluster?
- c. Are loop residues more widely distributed?
- d. How does the Ramachandran plot help you understand protein backbone conformation?
- e. Based on the Ramachandran plot, which type of secondary structure would you predict it is likely to belong to?